Server0

Physical | Config | Services | Desktop | Programming | Attributes

IP Configuration

IP Configuration

DHCP	Static
IPv4 Address	10.10.10.10
Subnet Mask	255.255.255.0
Default Gateway	10.10.10.2
DNS Server	0.0.0.0

DNS

DNS Service: On / Off

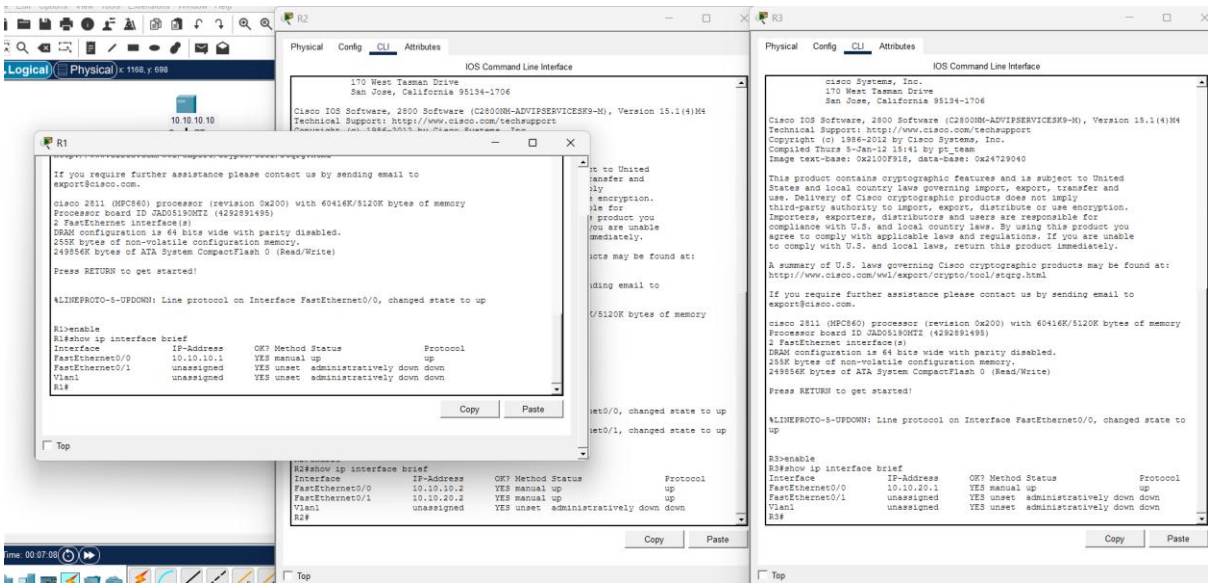
Resource Records

Name | Type: A Record

Address

Add | Save | Remove

No.	Name	Type	Detail
0	r1	A Record	10.10.10.1
1	r2	A Record	10.10.10.2
2	r3	A Record	10.10.20.1



R3

Physical Config CLI Attributes

IOS Cor

```
!
license udi pid CISCO2811/K9 sn FTX101
!
!
!
!
!
!
!
!
!
!
!
ip name-server 10.10.10.10
```

R1

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up


R1>enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
FastEthernet0/0        10.10.10.1      YES manual up                    up
FastEthernet0/1        unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/2/7 ms

R1#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

R1#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to down

R1#
R1#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#ping 10.10.10.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

R1#
```

R3

Physical Config CLI Attributes

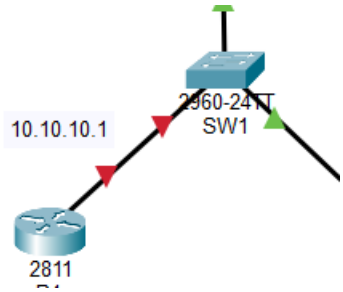
IOS Commanc

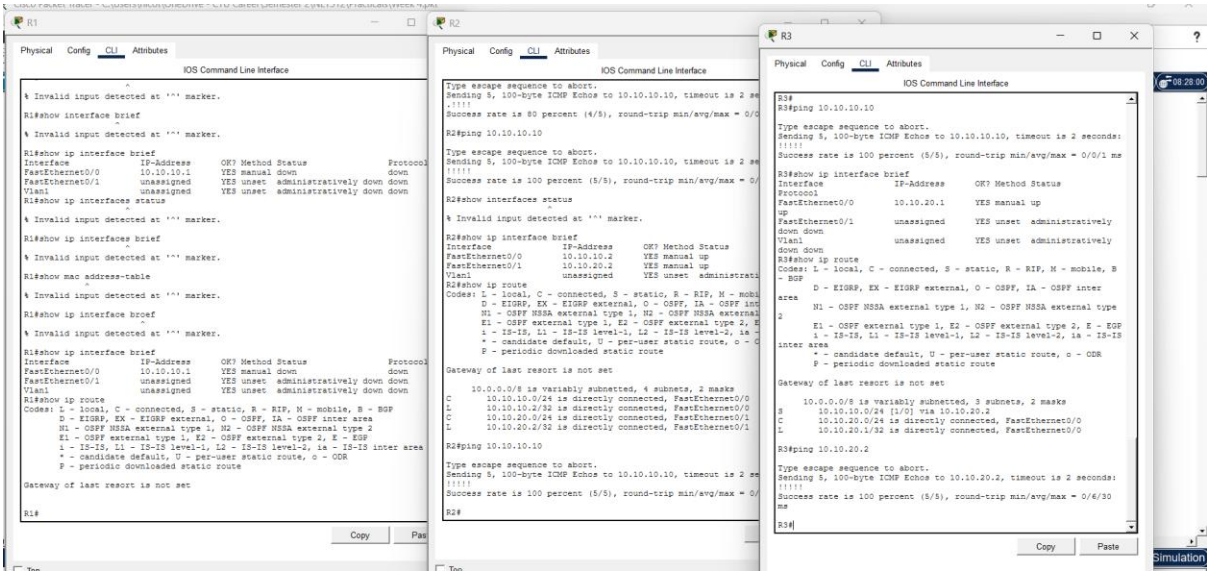
```
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 10.10.10.0 255.255.255.0 10.10.20
!
ip flow-export version 9
!
!
!
!
!
!
!
line con 0
!
line aux 0
!
line vty 0 4
 login
!
!
!
end

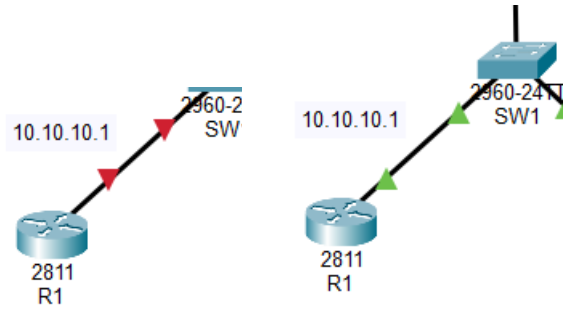

R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#
R3#ping 10.10.10.10

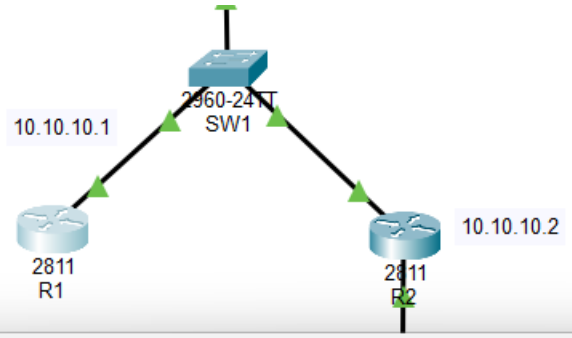
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10
!!!!!
Success rate is 100 percent (5/5), round-t

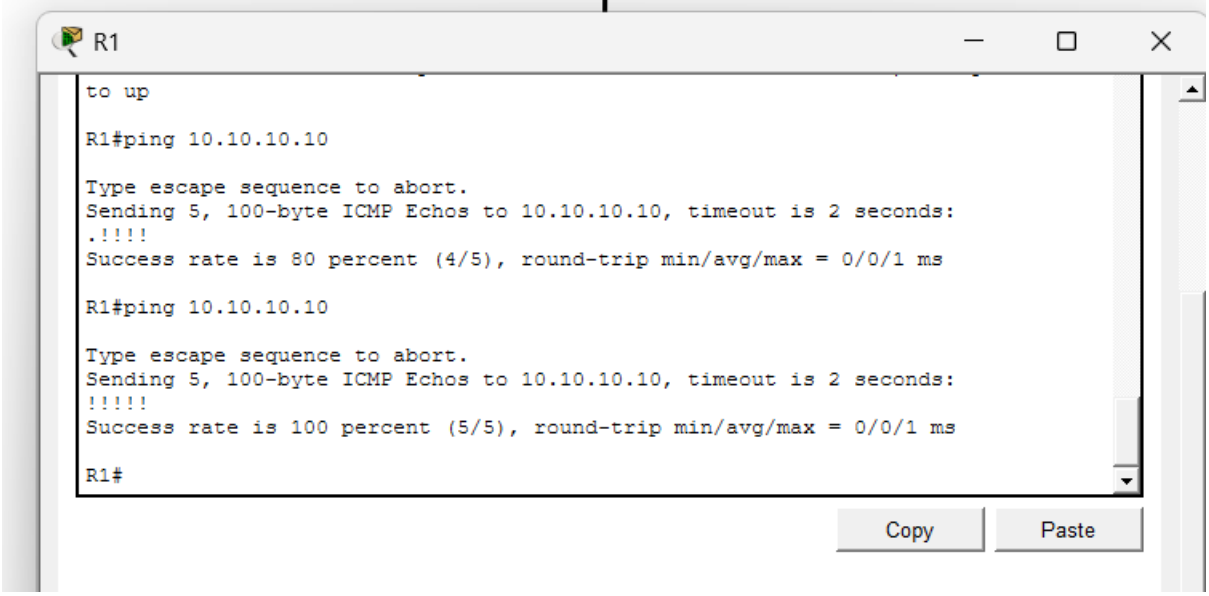
R3#
```




R1
Physical Config CLI Attributes
IOS Command Line Interface
% Invalid input detected at '^' marker.
R1#show interface brief
% Invalid input detected at '^' marker.
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.10.10.1 YES manual down down
FastEthernet0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R1#show ip interfaces status
% Invalid input detected at '^' marker.
R1#show ip interfaces brief
% Invalid input detected at '^' marker.
R1#show mac address-table
% Invalid input detected at '^' marker.
R1#show ip interface broef
% Invalid input detected at '^' marker.
R1#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 10.10.10.1 YES manual down down
FastEthernet0/1 unassigned YES unset administratively down down
Vlan1 unassigned YES unset administratively down down
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
R1#
Copy
R2
Physical Config CLI Attributes
IOS Command Line Interface
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
R2#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
R2#show interfaces status
% Invalid input detected at '^' marker.
R2#show ip interface brief
Interface IP-Address OK? Method Status
FastEthernet0/0 10.10.10.2 YES manual up
FastEthernet0/1 10.10.20.2 YES manual up
Vlan1 unassigned YES unset administrati
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobi
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF int
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external
E1 - OSPF external type 1, E2 - OSPF external type 2, E
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
* - candidate default, U - per-user static route, o - O
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C 10.10.10.0/24 is directly connected, FastEthernet0/0
L 10.10.10.2/32 is directly connected, FastEthernet0/0
C 10.10.20.0/24 is directly connected, FastEthernet0/1
L 10.10.20.2/32 is directly connected, FastEthernet0/1
R2#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 se
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/
R2#
R3
Physical Config CLI Attributes
IOS Command Line Interface
R3#
R3#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
R3#show ip interface brief
Interface IP-Address OK? Method Status
Protocol
FastEthernet0/0 10.10.20.1 YES manual up
up
FastEthernet0/1 unassigned YES unset administratively
down down
Vlan1 unassigned YES unset administratively
down down
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B
- BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter
area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type
2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS
inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route
Gateway of last resort is not set
10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
S 10.10.10.0/24 [1/0] via 10.10.20.2
C 10.10.20.0/24 is directly connected, FastEthernet0/0
L 10.10.20.1/32 is directly connected, FastEthernet0/0
R3#ping 10.10.20.2
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.20.2, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/6/30
ms
R3#
Copy
Paste
Simulation


2960-2
SW
10.10.10.1
2811
R1
10.10.10.1
2960-24
SW1
2811
R1


2960-24
SW1
10.10.10.1
2811
R1
2811
R2
10.10.10.2


R1
to up
R1#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/1 ms
R1#ping 10.10.10.10
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.10.10.10, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
R1#
Copy
Paste

The fault occurred at the physical because the cable was plugged out and put back into the wrong port on R1.

During troubleshooting, I checked the physical connections, switch interfaces, IP configurations and routing information. I used commands such as show interfaces status, show ip interface brief and show ip route to investigate the network.

The evidence that identified the problem was only R1 was having problems, but all the settings stayed the same.

I corrected the problem by checking what port it was plugged into, saw it was the wrong port, plugged it back into the right port and tested the pings again which came back positive. After applying the correction, I tested the connection again using ping and DNS lookup commands. The tests were successful, confirming that connectivity to the DNS server had been restored.